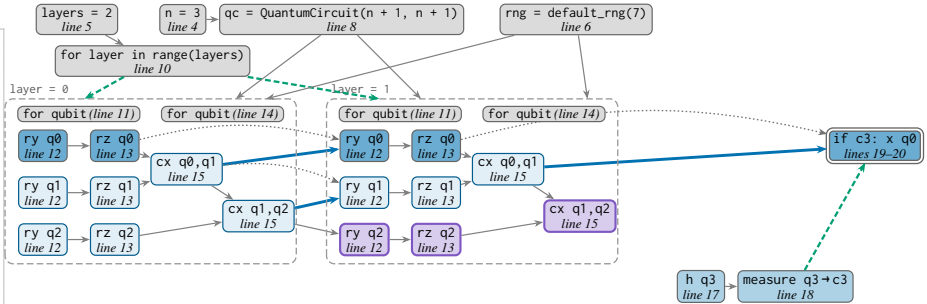
**(a) VQE ansatz**

```
 4 n = 3
 5 layers = 2
 6 rng = np.random.default_rng(7)
 7
 8 qc = QuantumCircuit(n + 1, n + 1)
 9
10 for layer in range(layers):
11     for qubit in range(n):
12         qc.ry(float(rng.uniform(0, np.pi)), qubit)
13         qc.rz(float(rng.uniform(0, 2 * np.pi)), qubit)
14     for qubit in range(n - 1):
15         qc.cx(qubit, qubit + 1)
16
17 qc.h(n)
18 qc.measure(n, n)
19 with qc.if_test((qc.clbits[n], 1)):
20     qc.x(0)
```

**(b) Dependence graph of the four rules, criterion $q_0$**




slices of $q_0$, of 19 operations: def-use 5, $F = 0.55$ · classical PDG 7, $F = 0.97$
QDG 16, exactly the dependences, $F = 1.00$ · QPDG 19, $F = 1.00$

data · quantum, not in the classical PDG · classical PDG only, skips a CNOT
control (measured bit or loop) · last write to the criterion · one iteration of a loop
kept by every rule, def-use included · added by the classical PDG · added by the QDG
added by the QPDG, not a dependence · classical statement in the QDG slice